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Face-to-Face Dimensions for Integral Flanged Globe-Style Control Valve Bodies (Classes 125, 150, 250, 300, and 600)

Approved 23 April 2007

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1 Scope

This standard applies to integral flanged globe-style control valves, sizes 15 mm (1/2 inch) through 400 mm (16 inches), having top, top and bottom, port, or cage guiding.

2 Purpose

The purpose of this standard is to aid users in their piping design by providing Class 125 flat face, and Classes 150, 250, 300, and 600 raised face, flanged control valve dimensions, without giving special consideration of the equipment manufacturer to be used.

3 Definitions

For definitions of terms used in this standard, see ANSI/ISA-75.05.01-2000 (R2005), Control Valve Terminology.

4 Dimensional data

4.1 Face-to-face dimensions for integral flanged globe-style control valves. See Table 1.

4.2 For end-to-end dimensions of flanged with ring joint ends, Dimension "A" shall be adjusted accordingly to ASME B16.10-2000 (R2003), paragraph 3.3.

Table 1 — Face-to-face dimensions for integral flanged globe-style control valves

Nominal valve Size		PN 20 (Classes 125 and 150)		PN 50 (Classes 250 and 300)		PN 110 (Class 600)		Tolerance	
		Dimension “A”		Dimension “A”		Dimension “A”			
mm	Inches	mm	Inches	mm	inches	mm	inches	mm	inches
15	½	184	7.25	190	7.50	203	8.00	±1.6	±0.062
20	¾	184	7.25	194	7.62	206	8.12	±1.6	±0.062
25	1	184	7.25	197	7.75	210	8.25	±1.6	±0.062
40	1-1/2	222	8.75	235	9.25	251	9.88	±1.6	±0.062
50	2	254	10.00	267	10.50	286	11.25	±1.6	±0.062
65	2-1/2	276	10.88	292	11.50	311	12.25	±1.6	±0.062
80	3	298	11.75	318	12.50	337	13.25	±1.6	±0.062
100	4	352	13.88	368	14.50	394	15.50	±1.6	±0.062
150	6	451	17.75	473	18.62	508	20.00	±1.6	±0.062
200	8	543	21.38	568	22.38	610	24.00	±1.6	±0.062
250	10	673	26.50	708	27.88	752	29.62	±1.6	±0.062
300	12	737	29.00	775	30.50	819	32.25	±3.2	±0.125
350	14	889	35.00	927	36.50	972	38.25	±3.2	±0.125
400	16	1016	40.00	1057	41.62	1108	43.62	±3.2	±0.125

ISO 7005-1 designates classes 150 and 300 as PN20 and PN50, respectively.
ISO 7005-2 designates classes 125 and 250 as PN20 and PN50, respectively.



NOTE 1 Class 125 and PN 20 grey cast iron flanges. Classes 150 and 300 cast copper alloy flanges.

NOTE 2 Classes 150, 300, and 600, and PN20, PN50, and PN110 cast steel/alloy steel flanges. Class 600 cast copper alloy flanges. Class 250 and PN50 grey cast iron flanges. Classes 150 and 300, and PN20 and PN50 cast ductile iron flanges.

5 References

The following bibliography is included for the definition of pressure classes and flange dimensions, material identification, and cross-reference information.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B16.1-1998	Cast Iron Pipe Flanges and Flanged Fittings, Classes 25, 125, and 250
ASME B16.5-2003	Pipe Flanges and Flanged Fittings: NPS 1/2 through 24
ASME B16.10-2000 (R2003)	Face-to-Face and End-to-End Dimensions of Valves
ASME B16.24-2001	Cast Copper Alloy Pipe Flanges and Flanged Fittings: Class 150, 300, 400, 600, 900, 1500, and 2500
ASME B16.34-2004	Valves Flanged, Threaded and Welding End
ASME B16.42-1998	Ductile Iron Pipe Flanges and Flanged Fittings: Classes 150 and 300
Available from:	ASME Three Park Avenue New York, NY 10016-5990 Tel: 800-843-2763 http://www.asme.org

INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC)

IEC 60534-3-1 (2000)	Industrial-Process Control Valves - Part 3-1: Dimensions - Face-to-Face Dimensions for Flanged, Two-Way, Globe-Type, Straight Pattern and Centre-to-Face Dimensions for Flanged, Two-Way, Globe-Type, Angle Pattern Control Valves
Available from:	IEC 3, rue de Varembé P.O. Box 131 CH - 1211 Geneva 20 Switzerland http://www.iec.ch

INTERNATIONAL STANDARDS ORGANIZATION (ISO)

ISO 7005-1:1992	Metallic Flanges – Part 1: Steel Flanges
ISO 7005-2:1988	Metallic Flanges – Part 2: Cast Iron Flanges

Available from:

ISO
1, rue de Varembé
P.O. Box 56
CH-1211 Geneva 20
Switzerland
http://www.iso.ch

Annex A — Dimensions for metrically sized valves

This annex is not part of ANSI/ISA-75.08.01-2002 (R2007), but it is included to facilitate its use.

The ASME standards listed in the bibliography contain valve body design information in addition to face-to-face dimensions. Dimensions for metrically sized valves are nominal conversions that are conventionally used in documents by the Manufacturers Standardization Society (MSS) of the Valve and Fitting Industry (MSS-SP-86-2002), by the International Organization for Standardization (ISO), and by the International Electrotechnical Commission (IEC).

Control valve initial dimensions, 10 inches - 16 inches, were listed in the Fluid Control Institute (FCI) Standard, FCI-65-2-1975, which has been withdrawn from publication.

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